

JALAL UDDIN MD AKBAR

Pekan, Malaysia

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Research Interests: Computer Vision, Deep Learning, Medical Image Analysis

EDUCATION

Universiti Malaysia Pahang Al-Sultan Abdullah, Malaysia Mar 2022 – Feb 2026

Master of Science in Soft Computing and Intelligent Systems

Viva Passed, Awaiting Conferral | Cumulative Marks: 88% 🔗

Thesis: Enhanced YOLOv8 with Attention Mechanism for Tomato Ripeness Detection in Smart Agriculture.

(Nominated for Best Thesis Award)

International Islamic University Chittagong, Bangladesh Mar 2016 – Aug 2021

Bachelor of Science in Computer Science and Engineering

Final grade: 3.23/4 🔗

EXPERIENCE

National Yang Ming Chiao Tung University, Taiwan Jul 2025 – Jan 2026
TEEP Research Intern Taipei, Taiwan

- Designed a GAN-based model to reconstruct high-fidelity PET images from 1% ultra-low-dose acquisitions.
- Optimized the architecture to balance denoising stability and texture hallucination, recovering sub-millimeter anatomical structures.
- Validated using ROI analysis and line profiles; achieved superior contrast recovery versus conventional CNN-based methods.

National Science and Technology Council (NSTC) Nov 2024 – Jan 2025
IIPP Research Fellow 🔗 Taipei, Taiwan

- Developed and optimized AI-driven models (U-Net and CBAM-U-Net) to enhance PET image quality for low-dose imaging.
- Integrated attention-based modules to strengthen feature extraction and improve reconstruction consistency.
- Focused on reducing radiation exposure while maintaining high-resolution imaging through deep learning enhancements.

Universiti Malaysia Pahang Al-Sultan Abdullah, Malaysia Jul 2022 – Jun 2025
Graduate Research Assistant Pekan, Malaysia

- Developed Modified-YOLO and Mask R-CNN pipelines for crop detection, achieving 97.4% mAP in greenhouse environments.
- Improved small-target recognition with Swin Transformer components, increasing accuracy by 4.6% and recall by 2.7%.
- Built real-time fruit maturity detection models, reducing inference time to 9.2 ms while maintaining 95.5% precision.
- Deployed a lightweight detector on NVIDIA Jetson devices, achieving 21.3 FPS on Xavier NX with 0.93 detection precision.

- Prepared research reports and supported manuscript preparation for journal submission.

Universiti Malaysia Pahang Al-Sultan Abdullah, Malaysia

Graduate Teaching Assistant 

Jan 2023 – Sep 2024

Pekan, Malaysia

- Conducted lab sessions for Network Programming and Algorithm & Complexity.
- Developed lab activities and evaluated student coursework.

PUBLICATIONS

Journal Articles

1. **J. U. M. Akbar**, S. F. Kamarulzaman, A. J. M. Muzahid, M. A. Rahman, M. Uddin, "A Comprehensive Review on Deep Learning Assisted Computer Vision Techniques for Smart Greenhouse Agriculture," *IEEE Access*, 2024. (WoS; IF 3.9) 
2. **J. U. M. Akbar** and S. F. Kamarulzaman, "YOLOv8s-Swin: Enhanced Tomato Ripeness Detection for Smart Agriculture," *IJACSA*, 16(8), 2025. 
3. Jumadi, J., & **Akbar, J. U. M.**, "Hybrid GRU-KAN Model for Energy Consumption Prediction in Commercial Building Cooling," *IJACI*, 1(2), 69–78, 2025. 
4. E. H. Tusher, M. A. Ismail, F. Khan, A. F. M. Raffei, **J. U. M. Akbar**, "Semi-Supervised Learning: Assisted Cardiovascular Disease Forecasting using Self-learning Approaches," *J. Adv. Res. Appl. Sci. Eng. Tech.*, 2024. (Scopus; Q2) 
5. M.D.M. Rizwan, M.A. Hossain, A.N. Zulkifili, **J. U. M. Akbar**, L. Singh, S. F. Kamarulzaman, "YOLO-FES: An Improved Elephant Intrusion Detector Based on YOLOv8n," *IEEE Access*, 2025. (WoS; IF 3.9) 

Conference Papers

1. **J. U. M. Akbar**, S. F. Kamarulzaman, E. H. Tusher, "Plant Stem Disease Detection Using Machine Learning Approaches," *ICCCNT*, 2023. 
2. M. S. Islam, M. M. Hasan, S. Abdullah, **J. U. M. Akbar**, N. H. M. Arafat, S. A. Murad, "A Deep Spatio-temporal Network for Vision-based Sexual Harassment Detection," *ETCCE*, 2021. 

Book Chapter

1. E. H. Tusher, **J. U. M. Akbar**, R. I. Rabbi, M. Hasan, "Energy Forecasting Utilizing CNN-LSTM Attention Mechanism: Empirical Evidence from the Spanish Electricity Market," in *Machine Learning Technologies on Energy Economics and Finance*, Springer, 2025. 

Under Review

- **J. U. M. Akbar**, S. F. Kamarulzaman, "YOLOv9-ResCBAM: Enhancing Tomato Ripeness Detection in Greenhouses through Advanced Object Detection," *Pertanika Journal of Science & Technology*, 2025. (Under review)

SELECTED COURSEWORK

Core CS Foundations: Discrete Mathematics; Computer Algorithms; Data Structures; Object Oriented Programming I & II; Research Methodology & Seminar; Database System.

Mathematics: Probability; Statistics; Linear Algebra and Calculus; Complex Analysis and Differential Equations, Numerical Methods; Adv. Probability and Stochastic Processes; Statistical Methods I/II;

Deep Learning / AI: Artificial Intelligence; Machine Learning and Data Mining ; Computer Graphics; Computer Simulation and Modeling; Neural Network and Fuzzy System .

TECHNICAL SKILLS

Programming Languages	Python (TensorFlow, Keras, PyTorch), MATLAB/Simulink, C/C++
Frameworks & Libraries	OpenCV, Sci-kit learn, NumPy, Pandas, SciPy, Matplotlib
Software Tools	Microsoft Office, Overleaf & LaTeX
Language Proficiency	English: Duolingo English Test  Overall Score: 125/160 (Upper Intermediate, CEFR B2) Speaking: 125 Listening: 120 Reading: 125 Writing: 125

HONORS AND AWARDS

TEEP Research Internship  <i>Taiwan Ministry of Education (MOE)</i> A 6-month research internship awarded by MOE.	Jul 2025
NSTC IIPP Fellowship  <i>National Science and Technology Council (NSTC), Taiwan</i> A 3-month research fellowship awarded by NSTC.	Sep 2024
Postgraduate Holistic Incentive Scheme  <i>Universiti Malaysia Pahang Al-Sultan Abdullah</i> Competitive scholarship awarded by UMPSA.	Mar 2024
Postgraduate Research Grants Scheme (PGRS) <i>Universiti Malaysia Pahang Al-Sultan Abdullah</i> Competitive research scholarship awarded by UMPSA.	May 2023
Master's Research Scheme (MRS)  <i>Universiti Malaysia Pahang Al-Sultan Abdullah</i> Merit-based scholarship awarded by the Institute of Postgraduate Studies (IPS).	Dec 2022

CERTIFICATIONS

• Huawei Certified ICT Associate – AI (HCIA-AI V3.0) 	Oct 2022
• Google IT Support Professional Certificate 	Dec 2020

LEADERSHIP & ACTIVITIES

• International PG Student Representative, UMPSA (Malaysia) <i>Represented PG students; coordinated support and student communication.</i>	Jan 2023 – Jan 2024
• Head of R&D, NextGen Leaders Program, CKH Network (Bangladesh) <i>Led R&D activities and program initiatives; supported participant projects.</i>	Jan 2021 – Apr 2022
• Executive Member, IIUC Computer Club (Bangladesh) <i>Organized technical activities and assisted with event coordination.</i>	Oct 2019 – Sep 2020
• Senior Rover Mate, IIUC Rover Scout Group (Bangladesh) <i>Led volunteering activities and team-based outdoor programs.</i>	Jun 2019 – Jul 2020

REFERENCES

Up to 3 references available upon request.